QUARTERLY PROGRESS REPORT

Progress Report for the Quarter July to September 2026

A month-by-month account of the work completed, the work in progress, and the plan for the next quarter.

SUBMITTED TO	CIC-TBI Cluster Innovation Centre, University of Delhi
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1 Executive Summary

Protopus is developing machines and accessories for hardware prototyping, guided by an AI engineer called Proto. The first machine, Baker, produces a finished printed circuit board on the desk. The second, Mixer, tests it. The aim is to make the first one to ten boards of a design faster and easier to produce, while volume manufacturing remains with fabrication houses.

During the quarter, we completed our research and launched the website (July). We finalised the prototype design and began building the first Baker (August). We then worked on integrating software with hardware, opened discussions for proof-of-concept funding, and documented costs, workflow and a manufacturing plan (September).

Highlights

- Research completed on the problem, users, competition and market.
- Website launched, with pages for both machines.
- Verbal commitment from Samuel Beek (Founder and CEO, Schematik) to fund Protopus after the proof of concept.
- Prototype design finalised: a 15-step process, machine architecture and subsystem specification.
- First prototype build and software–hardware integration under way.
- Discussions opened with STMicroelectronics for proof-of-concept funding.
- Costed bill of materials (70 items), cost report, workflow document and China manufacturing plan completed.

Status of milestones

Milestone	Month	Status
Research: problem, users, competition, market	July	COMPLETED
Brand identity and website launch	July	COMPLETED
Meeting with Samuel Beek, Schematik	July	COMPLETED
Prototype design finalised	August	COMPLETED
Build of the first Baker prototype	Aug onwards	IN PROGRESS
Software–hardware integration	Sep onwards	IN PROGRESS
Proof-of-concept funding, STMicroelectronics	Sep onwards	IN DISCUSSION
Probe Matrix accessory	September	CONCEPT STAGE
Cost report, workflow, manufacturing plan, pitch	September	COMPLETED

Table 1. Status of milestones as on 30 September 2026. Dates within each month are approximate.

MONTH 1

2 Progress in July 2026

July was devoted to understanding the problem, the people who face it and the market, and to taking Protopus to the public and to investors.

2.1 Problem study

Early July

COMPLETED

We mapped how a first circuit board is produced in India today. It requires a fabrication order, 7–20 days of shipping and customs, hand soldering or a paid assembly service, and separate instruments for testing. One assembled board from JLCPCB, landed in India, costs about ₹6,700 per attempt.

2.2 User study

Early July

COMPLETED

We divided hardware development into eight stages: specification, parts, schematic, layout, fabrication, assembly, bring-up and iteration. Each stage requires new knowledge and loses people. The first four are being addressed by AI design software. The last four are physical, and this is where we chose to work.

2.3 Competition study

Early to mid July

COMPLETED

We studied existing tools and their prices: desktop PCB mills (LPKF ProtoMat, Makera Carvera), the Bantam Tools mill (discontinued in 2025), Voltera's ink printer, Neoden pick-and-place machines, USB instruments such as the Digilent Analog Discovery 3, fabrication houses, and AI design tools such as Flux.ai. Each addresses only one stage, and none is driven by an AI.

2.4 Market study

Mid July

COMPLETED

We sized our first market in India: 5,868 AICTE engineering colleges, about 2.25 lakh DPIIT-recognised startups (of which we assume about 1% build hardware), and 10,000 Atal Tinkering Labs, with 50,000 more announced.

2.5
Mid July
COMPLETED

Product concept and positioning

We defined the system. Proto, an AI engineer, plans the build. Baker makes the board: milling, drilling, solder mask, stencil, paste and reflow. Mixer tests it. The machines are positioned for prototyping, the first one to ten boards of a design.

2.6
Mid to late July
COMPLETED

Brand identity and website launch

We created the name, logo and visual identity, and published the website with dedicated pages for Baker and Mixer (see Annexure A).

2.7
Late July
COMPLETED

Meeting with Samuel Beek, Founder and CEO of Schematik

We presented Protopus to Samuel Beek. He gave a verbal commitment to fund Protopus once the proof of concept is achieved. The commitment is not yet in writing and depends on the proof of concept.

MONTH 2

3 Progress in August 2026

In August we converted the concept into an engineering design, selected every part, and began building the first Baker prototype.

3.1 Baker process design

Early August

COMPLETED

We fixed the 15-step sequence that turns a bare copper board into a finished one. In order, the machine feeds the blank and probes its surface; mills the traces and drills the holes; cleans the board; coats, UV-images, develops and cures the solder mask; laser-cuts the stencil and applies solder paste; places the parts using a printed jig; reflows the board; and hands it over for test.

3.2 Machine architecture

Early to mid August

COMPLETED

A single moving shuttle carries the board through two process zones in one coordinate frame. Zone A mills, drills and laser-cuts the stencil. Zone B coats, images and cures the solder mask. A refill magazine feeds the next blank, so the board does not need loading each time. The placement jig is generated from the Gerber file and printed inside the machine.

3.3 Subsystem specification

Mid August

COMPLETED

Zone A uses a 500 W ER11 spindle with a touch probe, and a 10 W, 445 nm laser for the stencil. Zone B uses a syringe-dispensed liquid solder mask, imaged by a 10.1-inch, 405 nm LCD. Reflow uses a 300 W silicone hotplate with SSR control and a MAX31855 thermocouple, with low-temperature Sn42Bi58 solder (138 °C). Control is a Raspberry Pi 5 with a BTT Octopus Pro board running Klipper.

3.4

Mid August

COMPLETED

Mixer architecture, for a later stage

Mixer is defined as eight instruments on one cable: a 2-channel oscilloscope, a 2-channel waveform generator, a 16-channel logic analyser, a pattern generator, a protocol decoder, two programmable supplies and a 24-bit multimeter, built around a Tang Primer 20K FPGA. It is planned for after the Baker proof of concept.

3.5

Mid to late August

COMPLETED

Parts selection and bill of materials

We selected the parts for every Baker subsystem (motion, spindle and laser, mask and UV, reflow, control electronics and frame) and built the bill of materials, so that procurement could begin.

3.6

Late August

IN PROGRESS

Build of the first prototype

We began procurement and assembly of the first Baker prototype.

MONTH 3

4 Progress in September 2026

In September we connected the software to the machine, opened funding discussions for the proof of concept, conceptualised a new accessory, and documented the costs, workflow and manufacturing plan.

4.1 Baker Studio and Proto: the software layer

Through September

IN PROGRESS

We worked on Baker Studio, the software in which Proto checks a design and plans the machine job. Proto reads schematics and layouts through the Flux.ai API, flags problems such as traces that are too thin, proposes a fix for the user to accept, and plans the milling, drilling, mask, stencil and jig. Proto's conversational character is built on Gemini.

4.2 Software–hardware integration

Through September

IN PROGRESS

We worked on connecting Baker Studio to the prototype's control electronics through the Klipper firmware, with Moonraker and a job runner, so that a planned job can drive the machine.

4.3 Costed bill of materials and cost report

Mid to late September

COMPLETED

We costed the Baker bill of materials line by line (70 items). The first Baker prototype comes to ₹3,73,418, including import duty, courier and a 15% contingency. The Mixer prototype comes to ₹74,847. With parts sourced from China and assembly in Delhi at 100 units, production cost is projected at ₹1,44,842 for Baker and ₹13,740 for Mixer.

4.4 Workflow document

Late September

COMPLETED

We documented Baker's 15 steps, from a bare copper board to a reflowed board, followed by Mixer's test flow, as one continuous account of a single board.

4.5
Late September
COMPLETED

Manufacturing plan

We planned turnkey contract manufacturing in China, with testing, minimal assembly and shipping from India. The plan covers supplier selection, product-introduction gates, line stations, the test system, quality, landed cost, intellectual property, logistics and compliance.

4.6
Mid to late September
IN DISCUSSION

Proof-of-concept funding: STMicroelectronics

We approached STMicroelectronics to fund the prototype and the proof of concept. Discussions are ongoing; nothing has been committed yet.

4.7
Late September
CONCEPT STAGE

Probe Matrix conceptualised

Following the meeting with STMicroelectronics, we conceptualised the Probe Matrix: an accessory that places spring probes under the board, so that Proto can reach every net to test and program the board without manual probing. A concept page has been published (see Annexure A).

4.8
Late September
COMPLETED

Pitch and review

We prepared the pitch deck (problem, solution, competition, economics, plan and funding requirement) and received round-one review questions, for which written answers were prepared. As a result, the plan is now Baker first, with Mixer and the Probe Matrix after the proof of concept, and funding released in milestone-based instalments.

5 Outputs and Deliverables

The quarter produced the following measurable outputs and completed deliverables.

Output	Value
Baker process steps designed	15
Items in the costed Baker bill of materials	70
Machines designed / accessories conceived	2 / 1
First Baker prototype, costed build (incl. duty and 15% contingency)	₹3,73,418
Baker production cost, projected at 100 units	₹1,44,842
Deliverables completed	7

Table 2. Key outputs of the quarter. Costs are projections from our cost report.

Deliverable	Contents	Month
Website	Product website with pages for Baker and Mixer	July
Prototype design	Baker 15-step process, machine layout, subsystem specification; Mixer architecture	August
Bill of materials and costing workbook	Prototype BOM, cost breakup, and production BOM and unit cost for both machines	September
Cost report	Prototype and production cost for both machines, with duty and contingency	September
Workflow document	Baker’s 15 steps and Mixer’s test flow, step by step	September
Manufacturing plan	Turnkey contract manufacturing in China; testing and minimal assembly in India	September
Pitch deck	Problem, solution, competition, economics, plan and funding requirement	September
Probe Matrix concept page	Public description of the accessory concept	September

Table 3. Deliverables completed during the quarter.

6 Funding and Partnerships

The position of our funding discussions at the end of the quarter is as follows.

Party	Nature	Status
Samuel Beek, Founder and CEO, Schematik	Commitment to fund after the proof of concept	VERBAL COMMITMENT
STMicroelectronics	Discussions for funding of the prototype and proof of concept	IN DISCUSSION

Table 4. Funding discussions as on 30 September 2026.

Proposed funding for the proof of concept

We are seeking ₹15 lakh for a nine-month proof of concept, with no founder salary drawn. We propose that it is released in three instalments, each only after the previous milestone has been demonstrated. A lean version of ₹10 lakh would follow the same milestones.

Instalment	Amount	Released	Milestone that follows
1	₹6,00,000	At signing	Baker runs all 15 steps on one board (month 2)
2	₹5,00,000	After milestone 1	Process tests pass; Proto produces a Baker job from a plain-English brief; Baker Rev B built (month 5)
3	₹4,00,000	After milestone 2	Proof of concept: 20 boards made with Proto and Baker (month 6), followed by the seed round
Total	**₹15,00,000**		

Table 5. Proposed milestone-based instalments.

OCTOBER TO DECEMBER 2026

7 Plan for the Next Quarter

The next quarter is focused on completing the first Baker prototype and closing the funding for the proof of concept. Mixer and the Probe Matrix will remain at design stage until the proof of concept is achieved.

Planned activity	Target
Complete the build of the first Baker prototype	October–November
Run all 15 Baker steps end to end on one board	November–December
Continue software–hardware integration; Proto from a plain-English brief to Gerbers to a Baker job	November–December
Close proof-of-concept funding (STMicroelectronics and others)	October–December

Table 6. Planned activities, October to December 2026.

A Annexure: References and Notes

Reference	Link
Website	https://protopus-kappa.vercel.app/
Baker	https://protopus-kappa.vercel.app/#/baker
Mixer	https://protopus-kappa.vercel.app/#/mixer
Probe Matrix concept	https://probe-matrix.vercel.app/

Table A1. Online references.

Notes

- Dates within each month are approximate (early, mid or late).
- Costs and prices are projections from our cost report and have not been validated by pilots.
- Status labels: **Completed**, the work is done; **In progress**, the work has started and continues; **In discussion**, talks are under way and nothing is committed; **Concept stage**, the idea is defined but not built; **Verbal commitment**, agreed in conversation, not yet in writing.

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